1 Analysis of Variance and Designed Experiments

1.1 One-Way Layouts and the F Test

Analysis of variance compares several mean levels by splitting total variation into variation within treatment groups and variation between treatment groups. The method is a multisample continuation of the equal-variance two-sample procedure in Subsection **??**, with the design cautions from Principle **??** still in force.

Definition 1.1.1. *Balanced One-Way Layout*
For $I, J \in \mathbb{N}$, one factor has treatment levels $i = 1, \dots, I$ and each level has independent observations Y_{ij}, $j = 1, \dots, J$. The normal fixed-effects model satisfies $Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}$, $\sum_{i=1}^{I} \alpha_i = 0$, and independent errors $\varepsilon_{ij} \sim N(0, \sigma^2)$. *Structural Commentary*: $\mathbb{E}Y_{ij} = \mu + \alpha_i$, and $\sum_{i=1}^{I} \alpha_i = 0$ identifies μ as the average treatment mean. *Key Insight*: $H_0 : \alpha_i = 0 \ \forall i$ iff all treatment means are equal.

Definition 1.1.2. *One-Way Sums of Squares*
For a balanced one-way layout, set $\overline{Y}_{i\cdot} := \frac{1}{J}\sum_{j=1}^{J} Y_{ij}$ and $\overline{Y}_{\cdot\cdot} := \frac{1}{IJ}\sum_{i=1}^{I}\sum_{j=1}^{J} Y_{ij}$.

(i) *Total*: $SS_{\text{tot}} := \sum_{i=1}^{I}\sum_{j=1}^{J}(Y_{ij} - \overline{Y}_{\cdot\cdot})^2$.
(ii) *Within-treatment*: $SS_W := \sum_{i=1}^{I}\sum_{j=1}^{J}(Y_{ij} - \overline{Y}_{i\cdot})^2$.
(iii) *Between-treatment*: $SS_B := J\sum_{i=1}^{I}(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2$.

Proposition 1.1.3. *One-Way Sum-of-Squares Decomposition*
In Definition 1.1.2,

$$SS_{\text{tot}} = SS_W + SS_B.$$

Proof. Use

$$Y_{ij} - \overline{Y}_{\cdot\cdot} = (Y_{ij} - \overline{Y}_{i\cdot}) + (\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})$$

and square. The cross term vanishes because

$$\sum_{j=1}^{J}(Y_{ij} - \overline{Y}_{i\cdot}) = 0$$

for each treatment i. □

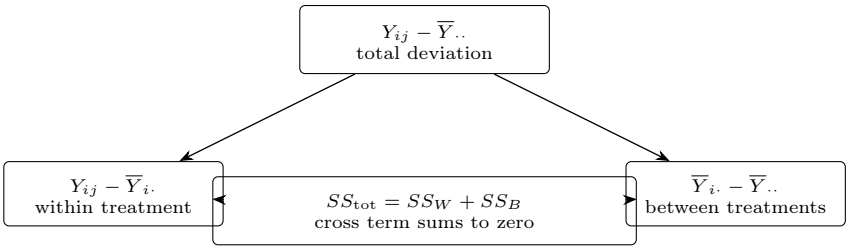


Figure 1: One-way analysis of variance separates total deviations into within-treatment residual variation and between-treatment mean variation.

Lemma 1.1.4. *Expected Squared Deviation from a Sample Mean*
Let $X_1, \dots, X_n$ be independent random variables with $\mathbb{E}X_i = \mu_i$ and common variance σ^2. If

$$\overline{X} := \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \overline{\mu} := \frac{1}{n}\sum_{i=1}^{n} \mu_i,$$

then

$$\mathbb{E}(X_i - \overline{X})^2 = (\mu_i - \overline{\mu})^2 + \frac{n-1}{n}\sigma^2.$$

Proof. Use $\mathbb{E}U^2 = (\mathbb{E}U)^2 + \operatorname{Var}(U)$ with $U = X_i - \overline{X}$. Independence gives

$$\operatorname{Var}(X_i - \overline{X}) = \sigma^2 + \frac{\sigma^2}{n} - 2\frac{\sigma^2}{n} = \frac{n-1}{n}\sigma^2.$$

□

Theorem 1.1.5. *Expected One-Way Sums of Squares*
Under Definition 1.1.1,

$$\mathbb{E}(SS_W) = I(J-1)\sigma^2, \qquad \mathbb{E}(SS_B) = (I-1)\sigma^2 + J\sum_{i=1}^{I}\alpha_i^2.$$

Thus

$$S_p^2 := \frac{SS_W}{I(J-1)}$$

is an unbiased estimator of the common error variance σ^2.

Proof. For SS_W, apply Lemma 1.1.4 inside each treatment group, where all observations have the same mean. For SS_B, apply the same lemma to the treatment means $\overline{Y}_{i\cdot}$, whose common error variance is σ^2/J and whose expectations are $\mu + \alpha_i$. □

Theorem 1.1.6. *Normal-Theory One-Way F Test*
Under Definition 1.1.1,

$$\frac{SS_W}{\sigma^2} \sim \chi^2_{I(J-1)}.$$

Under $H_0 : \alpha_1 = \cdots = \alpha_I = 0$,

$$\frac{SS_B}{\sigma^2} \sim \chi^2_{I-1}, \qquad SS_B \perp\!\!\!\perp SS_W.$$

Therefore the statistic

$$F = \frac{SS_B/(I-1)}{SS_W/[I(J-1)]}$$

has null distribution $F_{I-1,I(J-1)}$.

Proof. The within-group statement follows by applying Theorem **??** separately in each treatment and summing independent chi-square variables. Under H_0, the centered treatment means form the normal mean-residual part orthogonal to the within-treatment residuals, so the chi-square law and independence follow as in Theorem **??**. The final display is the defining ratio in Definition **??**. □

Method 1.1.7. *Unequal Sample Sizes*
If treatment i has J_i observations and $N = \sum_i J_i$, use

$$\overline{Y}_{i\cdot} := \frac{1}{J_i}\sum_{j=1}^{J_i} Y_{ij}, \qquad \overline{Y}_{\cdot\cdot} := \frac{1}{N}\sum_{i=1}^{I}\sum_{j=1}^{J_i} Y_{ij},$$

$$SS_W = \sum_{i=1}^{I}\sum_{j=1}^{J_i}(Y_{ij} - \overline{Y}_{i\cdot})^2, \qquad SS_B = \sum_{i=1}^{I} J_i(\overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot})^2.$$

The same algebra gives $SS_{\text{tot}} = SS_W + SS_B$, with degrees of freedom $N - I$ and $I - 1$. Under the usual normality, independence, and common-variance assumptions, the null statistic

$$F = \frac{SS_B/(I-1)}{SS_W/(N-I)}$$

is calibrated by $F_{I-1,N-I}$.

Example 1.1.8. *Seven Laboratories Measuring Tablets*
The source compares $I = 7$ laboratories, each with $J = 10$ determinations of chlorpheniramine maleate in tablets. Parallel boxplots, as in Method **??**, suggested differences in both center and spread. The reported analysis of variance table was

Source	*df*	*SS*	*MS*	*F*
Labs	6	.125	.021	5.66
Error	63	.231	.0037	
Total	69	.356		

The F statistic has 6 and 63 degrees of freedom; the source reports a tail probability below .01. The conclusion is evidence of systematic differences among laboratory mean levels, not merely random within-laboratory variation. The accompanying normal probability plot of residuals showed some lower-tail deviation but not a gross failure of normality.

Remark 1.1.9. *Assumptions and Robustness*
The one-way F test assumes independent errors, normal error shape, and common variance. Normality is mainly a small-sample calibration issue; equal sample sizes make the F test less sensitive to moderate variance

differences. Independence is more structural: if measurements are clustered, paired, blocked, or repeated on the same unit, the one-way independent-sample model is not the right owner of the dependence.

1.2 Multiple Comparisons After ANOVA

The one-way F test asks whether all treatment means are equal. Once it rejects, the scientific question often becomes local: which means differ, and by how much? The difficulty is simultaneous error control, already visible in Method **??**.

Definition 1.2.1. *Studentized Range Quantile*
Let $Q_{I,\nu}$ denote a random variable with the studentized range distribution for I treatment means and ν error degrees of freedom. $q_{I,\nu}(\alpha) := \inf\{q \in \mathbb{R} : \mathbb{P}(Q_{I,\nu} > q) \le \alpha\}$. *Structural Commentary*: In a balanced one-way layout, Tukey comparisons use the pooled error scale S_p from Theorem 1.1.5.

Method 1.2.2. *Tukey Pairwise Comparisons*
In a balanced normal one-way layout, simultaneous $100(1-\alpha)\%$ confidence intervals for all pairwise differences $\mu_{i_1} - \mu_{i_2}$ are

$$(\overline{Y}_{i_1\cdot} - \overline{Y}_{i_2\cdot}) \pm q_{I,I(J-1)}(\alpha)\frac{S_p}{\sqrt{J}}.$$

Equivalently, reject the pairwise null $\mu_{i_1} = \mu_{i_2}$ when

$$\left|\overline{Y}_{i_1\cdot} - \overline{Y}_{i_2\cdot}\right| > q_{I,I(J-1)}(\alpha)\frac{S_p}{\sqrt{J}}.$$

The protection is simultaneous over all pairs, so the threshold is larger than the threshold from a single pooled two-sample t comparison.

Example 1.2.3. *Tablet Laboratories: Tukey Comparisons*
For the tablet laboratory data in Example 1.1.8, the reported ordered sample means were

Lab	1	3	7	2	5	6	4
Mean	4.062	4.003	3.998	3.997	3.957	3.955	3.920

Using $S_p = .06$ and the source's approximation $q_{7,60}(.05) = 4.31$, the Tukey half-width is about

$$4.31\frac{.06}{\sqrt{10}} = .082.$$

Thus lab 1 is significantly higher than labs $4, 5, 6$, and lab 3 is significantly higher than lab 4; the remaining pairwise differences are not significant at the simultaneous 5% level.

Method 1.2.4. *Bonferroni Pairwise Comparisons*
Suppose k pairwise hypotheses or confidence intervals are planned. Testing each individual hypothesis at level α/k, or forming each individual confidence interval at level $1-\alpha/k$, controls the family-wise error rate at most α. For pairwise mean comparisons using the one-way pooled error variance, a two-sided Bonferroni interval has the form

$$(\overline{Y}_{i_1\cdot} - \overline{Y}_{i_2\cdot}) \pm t_{\nu,1-\alpha/(2k)} S_p \sqrt{\frac{1}{J_{i_1}} + \frac{1}{J_{i_2}}},$$

with $\nu = N - I$ in the unequal-size case. Unlike Tukey's balanced formula, this method adapts directly to unequal treatment sizes and to selected planned comparisons.

Example 1.2.5. *Tablet Laboratories: Bonferroni Comparisons*
For all $\binom{7}{2} = 21$ pairwise laboratory comparisons, the source uses a Bonferroni calculation with a critical value near 3.16. The resulting simultaneous 95% half-width is about

$$.085,$$

very close to the Tukey half-width .082 in Example 1.2.3. It again identifies lab 1 as significantly higher than labs $4, 5, 6$.

Remark 1.2.6. *The Global Test Does Not Name the Difference*
Rejecting the one-way ANOVA null says that not all treatment means are equal. It does not identify the responsible pairs or estimate their signs with simultaneous reliability. Planned comparisons, Tukey intervals, and Bonferroni intervals answer different questions and should be chosen according to the comparison family that was scientifically meaningful before looking at the data.

1.3 Rank-Based One-Way ANOVA

When normality or outlier sensitivity is a concern, a one-way comparison can be made after replacing observations by pooled ranks. This continues the rank-test philosophy from Subsection **??**.

Definition 1.3.1. *Kruskal–Wallis Ranks*
In an unequal-size one-way layout, treatment i has observations Y_{ij} for $j = 1, \dots, J_i$, and $N := \sum_{i=1}^{I} J_i$. Rank the pooled sample $\{Y_{ij} : 1 \le i \le I,\ 1 \le j \le J_i\}$ increasingly, with tied observations assigned the average of occupied ranks.

(i) *Pooled rank*: $R_{ij} := \mathrm{rank}_{\mathrm{pool}}(Y_{ij})$.
(ii) *Treatment mean rank*: $\overline{R}_{i\cdot} := \frac{1}{J_i} \sum_{j=1}^{J_i} R_{ij}$.
(iii) *Grand mean rank*: $\overline{R}_{\cdot\cdot} := \frac{N+1}{2}$.

Method 1.3.2. *Kruskal–Wallis Test*
Define the rank between-group sum of squares

$$SS_B^{(R)} := \sum_{i=1}^{I} J_i (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2.$$

The Kruskal–Wallis statistic is

$$K = \frac{12}{N(N+1)} SS_B^{(R)} = \frac{12}{N(N+1)} \left(\sum_{i=1}^{I} J_i \overline{R}_{i\cdot}^{\,2} \right) - 3(N+1).$$

The exact null law is obtained by enumerating rank assignments. For ordinary moderate sample sizes, the null law is well approximated by χ^2_{I-1}.

Example 1.3.3. *Tablet Laboratories by Ranks*
For the tablet laboratory data of Example 1.1.8, the source reports

$$K = 29.51.$$

Using the χ^2_6 approximation, the tail probability is below .005. Thus the rank analysis, like the normal-theory F test, gives strong evidence that the laboratories do not all have the same response distribution.

Corollary 1.3.4. *Two Groups Recover Mann–Whitney*
When $I = 2$, the Kruskal–Wallis statistic is a monotone function of the rank sum for either group. Therefore the two-group Kruskal–Wallis test is equivalent to the Mann–Whitney rank-sum test of Method **??**.

Proof. With two groups, the two rank sums determine each other because their total is $N(N+1)/2$. The statistic $SS_B^{(R)}$ therefore depends on only one rank sum, and it is largest exactly when that rank sum is farthest from its null center. □

Method 1.3.5. *Rank-Based Follow-Up Comparisons*
After a Kruskal–Wallis rejection, pairwise Mann–Whitney tests may be used as follow-up comparisons. To protect the family of decisions, apply the Bonferroni control from Method 1.2.4 to the chosen pairwise rank tests. This is especially natural when only a small set of scientifically planned comparisons is of interest.

Remark 1.3.6. *What Rank ANOVA Tests*
The Kruskal–Wallis null is equality of the treatment response distributions, not merely equality of means. It is often interpreted as a location comparison when the group distributions have similar shapes. Because it uses ranks, it makes no normality assumption and is less driven by extreme numerical values than the one-way F test.

1.4 Two-Way Layouts and Interaction

A two-way layout studies two factors at once. It estimates main effects for each factor and checks whether the effect of one factor changes across levels of the other.

Definition 1.4.1. *Two-Way Additive Parametrization*
For one observation Y_{ij} at factor levels $i = 1, \dots, I$ and $j = 1, \dots, J$, set $\overline{Y}_{i\cdot} := J^{-1} \sum_{j=1}^{J} Y_{ij}$, $\overline{Y}_{\cdot j} := I^{-1} \sum_{i=1}^{I} Y_{ij}$, and $\overline{Y}_{\cdot\cdot} := (IJ)^{-1} \sum_{i=1}^{I} \sum_{j=1}^{J} Y_{ij}$.

(i) *Intercept*: $\widehat{\mu} := \overline{Y}_{\cdot\cdot}$.
(ii) *Factor A coefficient*: $\widehat{\alpha}_i := \overline{Y}_{i\cdot} - \overline{Y}_{\cdot\cdot}$, $1 \le i \le I$.
(iii) *Factor B coefficient*: $\widehat{\beta}_j := \overline{Y}_{\cdot j} - \overline{Y}_{\cdot\cdot}$, $1 \le j \le J$.

(iv) *Additive fitted value*: $\widehat{Y}_{ij} := \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j$.
(v) *Interaction residual*: $\widehat{\delta}_{ij} := Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}_{\cdot\cdot}$, so $Y_{ij} = \widehat{\mu} + \widehat{\alpha}_i + \widehat{\beta}_j + \widehat{\delta}_{ij}$.

Structural Commentary: In exact arithmetic, $\sum_{i=1}^{I} \widehat{\alpha}_i = \sum_{j=1}^{J} \widehat{\beta}_j = 0$, $\sum_{i=1}^{I} \widehat{\delta}_{ij} = 0\ \forall j$, and $\sum_{j=1}^{J} \widehat{\delta}_{ij} = 0\ \forall i$; displayed rounded data may violate these identities numerically.

Remark 1.4.2. *Interaction and Parallel Profiles*
In an additive model, the fitted difference between two levels of one factor is the same at every level of the other factor. Thus profile lines are parallel. Visible nonparallelism is evidence of interaction: the comparison between levels depends on the background level of the other factor.

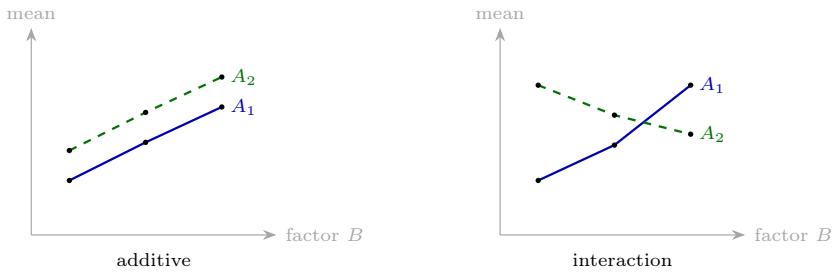


Figure 2: Parallel profiles represent an additive two-way model; changing separation between profiles indicates interaction.

Definition 1.4.3. *Balanced Two-Way Normal Model*
For factor levels $i = 1, \dots, I$ and $j = 1, \dots, J$ with $K > 1$ independent observations per cell, the fixed-effects model satisfies $Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \varepsilon_{ijk}$, independent errors $\varepsilon_{ijk} \sim N(0, \sigma^2)$, $\sum_{i=1}^{I} \alpha_i = 0$, $\sum_{j=1}^{J} \beta_j = 0$, $\sum_{i=1}^{I} \delta_{ij} = 0\ \forall j$, and $\sum_{j=1}^{J} \delta_{ij} = 0\ \forall i$. *Key Insight*: $\delta_{ij} = 0\ \forall (i,j)$ iff the cell means are additive in the two factors.

Definition 1.4.4. *Two-Way Sums of Squares*
For the model in Definition 1.4.3, write $\sum_i$, $\sum_j$, and $\sum_k$ for sums over $1, \dots, I$, $1, \dots, J$, and $1, \dots, K$.
$\overline{Y}_{ij\cdot} := K^{-1} \sum_k Y_{ijk}$.
$\overline{Y}_{i\cdot\cdot} := (JK)^{-1} \sum_j \sum_k Y_{ijk}$.
$\overline{Y}_{\cdot j\cdot} := (IK)^{-1} \sum_i \sum_k Y_{ijk}$.
$\overline{Y}_{\cdots} := (IJK)^{-1} \sum_i \sum_j \sum_k Y_{ijk}$.

(i) *Factor A*: $SS_A := JK \sum_i (\overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdots})^2$.
(ii) *Factor B*: $SS_B := IK \sum_j (\overline{Y}_{\cdot j\cdot} - \overline{Y}_{\cdots})^2$.
(iii) *Interaction*: $SS_{AB} := K \sum_i \sum_j (\overline{Y}_{ij\cdot} - \overline{Y}_{i\cdot\cdot} - \overline{Y}_{\cdot j\cdot} + \overline{Y}_{\cdots})^2$.
(iv) *Error*: $SS_E := \sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y}_{ij\cdot})^2$.
(v) *Total*: $SS_{\text{tot}} := \sum_i \sum_j \sum_k (Y_{ijk} - \overline{Y}_{\cdots})^2$.

Structural Commentary: $SS_{\text{tot}} = SS_A + SS_B + SS_{AB} + SS_E$.

Theorem 1.4.5. *Expected Two-Way Sums of Squares*
Under Definition 1.4.3,

$$\mathbb{E}(SS_A) = (I-1)\sigma^2 + JK \sum_{i=1}^{I} \alpha_i^2,$$

$$\mathbb{E}(SS_B) = (J-1)\sigma^2 + IK \sum_{j=1}^{J} \beta_j^2,$$

$$\mathbb{E}(SS_{AB}) = (I-1)(J-1)\sigma^2 + K \sum_{i=1}^{I} \sum_{j=1}^{J} \delta_{ij}^2,$$

$$\mathbb{E}(SS_E) = IJ(K-1)\sigma^2.$$

Proof. Apply Lemma 1.1.4 to the relevant marginal means and cell means. The parameter constraints remove mixed cross terms when the squared effects are summed. □

Theorem 1.4.6. *Two-Way Normal-Theory F Tests*
Under normality, the sums of squares in Definition 1.4.4 are independent. Also

$$\frac{SS_E}{\sigma^2} \sim \chi^2_{IJ(K-1)}.$$

Under the corresponding null hypotheses,

$$\frac{SS_A}{\sigma^2} \sim \chi^2_{I-1}, \qquad \frac{SS_B}{\sigma^2} \sim \chi^2_{J-1}, \qquad \frac{SS_{AB}}{\sigma^2} \sim \chi^2_{(I-1)(J-1)}.$$

Consequently, main effects and interaction are tested by

$$F_A = \frac{SS_A/(I-1)}{SS_E/[IJ(K-1)]}, F_B = \frac{SS_B/(J-1)}{SS_E/[IJ(K-1)]},$$

$$F_{AB} = \frac{SS_{AB}/[(I-1)(J-1)]}{SS_E/[IJ(K-1)]}.$$

Example 1.4.7. *Iron Retention: Transformation, Main Effects, and Interaction*
The source analyzes iron-retention data for two forms of iron and three dosage levels, with $K = 18$ observations per cell. Boxplots and cell standard deviations indicated that a logarithmic transformation stabilized spread. On the log scale, the reported ANOVA table was

Source	*df*	*SS*	*MS*	*F*
Iron form	1	2.074	2.074	5.99
Dosage	2	15.588	7.794	22.53
Interaction	2	.810	.405	1.17
Error	102	35.296	.346	
Total	107	53.768		

The iron-form effect has reported $p < .025$. The estimated log difference was .28, corresponding to a multiplicative factor $e^{.28} = 1.32$ on the original scale, with reported approximate confidence interval $(1.06, 1.65)$. The interaction test had reported $p = .31$, so the visible departure from parallelism in the source plot was not strong evidence that the iron-form comparison depended on dosage.

Remark 1.4.8. *Transformations and Diagnostics*
Two-way ANOVA inherits the normal-theory assumptions of the one-way model. Residual plots, normal probability plots, profile plots, and plots of cell spread against cell mean help decide whether a transformation is needed. A transformation can stabilize variance and improve symmetry, but it also changes additive effects on the transformed scale into multiplicative or otherwise nonlinear effects on the original scale.

1.5 Randomized Blocks and Friedman's Test

Randomized block designs compare treatments within relatively homogeneous blocks. Blocking removes a major source of background variation from the treatment comparison, much as pairing removes shared variation in Proposition **??**.

Definition 1.5.1. *Randomized Block Design*
For I treatments and J blocks, each block contains I experimental units and receives a within-block random permutation of the treatments. The additive block model satisfies $Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, $\sum_{i=1}^{I} \alpha_i = 0$, and $\sum_{j=1}^{J} \beta_j = 0$. *Key Insight*: $\alpha_i - \alpha_{i'}$ compares treatments after conditioning on block level j, while β_j absorbs block heterogeneity.

Method 1.5.2. *Randomized Block F Test*
Treat the randomized block model as a two-way layout with one observation per cell and no interaction. Let MS_A, MS_B, and MS_{AB} be the treatment, block, and residual mean squares. When the no-interaction model is adequate,

$$\mathbb{E}(MS_A) = \sigma^2 + \frac{J}{I-1}\sum_{i=1}^{I} \alpha_i^2, \qquad \mathbb{E}(MS_B) = \sigma^2 + \frac{I}{J-1}\sum_{j=1}^{J} \beta_j^2,$$

$$\mathbb{E}(MS_{AB}) = \sigma^2.$$

Thus the treatment null is tested by

$$F = \frac{MS_A}{MS_{AB}},$$

with $I-1$ and $(I-1)(J-1)$ degrees of freedom under the normal additive model.

Remark 1.5.3. *Interaction in a Block Design*
If treatment effects vary substantially across blocks, then MS_{AB} estimates σ^2 plus interaction variation. Using it as the denominator for the treatment test can make the test conservative, because the denominator is inflated by structure not included in the additive block model.

Example 1.5.4. *Itching Treatments with Subjects as Blocks*
In the source experiment, seven treatments for itching were compared on ten subjects, each subject receiving one treatment per day in randomized order. Subjects are blocks. The reported ANOVA table was

Source	*df*	*SS*	*MS*	*F*
Drugs	6	53013	8835	2.85
Subjects	9	103280	11476	3.71
Interaction	54	167130	3095	
Total	69	323422		

The drug statistic $F = 2.85$ on 6 and 54 degrees of freedom has reported $p < .025$, giving evidence of treatment differences. The subject effect is large but is not the experimental target; blocking prevents that subject-to-subject variation from being mistaken for treatment variation.

Method 1.5.5. *Friedman's Rank Test for Blocks*
For a nonparametric block analysis, rank the I treatment responses within each of the J blocks. Let $\overline{R}_{i\cdot}$ be the average rank of treatment i across blocks and let

$$\overline{R}_{\cdot\cdot} = \frac{I+1}{2}.$$

The Friedman statistic is

$$Q = \frac{12J}{I(I+1)} \sum_{i=1}^{I} (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2.$$

For sample sizes like the source itching experiment, the null distribution is well approximated by χ^2_{I-1}. Average ranks are assigned to ties.

Example 1.5.6. *Itching Treatments by Within-Subject Ranks*
For the itching data, the source reports average treatment ranks for no drug, placebo, papaverine, morphine, aminophylline, pentobarbital, and tripelennamine:

Treatment	No	Plac.	Pap.	Morph.	Amino.	Pento.	Trip.
Average rank	5.10	4.90	2.30	3.50	3.05	4.90	4.25

With $\overline{R}_{\cdot\cdot} = 4$, the source gives

$$\sum_{i} (\overline{R}_{i\cdot} - \overline{R}_{\cdot\cdot})^2 = 6.935, \qquad Q = 14.86.$$

Using the χ^2_6 approximation, the reported tail probability is below .025. The rank analysis reaches the same qualitative conclusion as the parametric block ANOVA.

Corollary 1.5.7. *Two Treatments Recover the Sign Test*
With two treatments in each block, Friedman's test uses only whether treatment 1 *or treatment* 2 *receives the larger within-block rank. Thus it carries the same information as the paired sign test from Example* **??**.

Proof. Within each block, the ranks are 1 and 2. The average rank of one treatment is therefore determined by the number of blocks in which that treatment has the larger response, which is exactly the sign-test count after ties are set aside or averaged according to the chosen convention. □

1.6 Exercise-Derived Extensions

The source exercises emphasize that ANOVA is a language for decomposing variation, not just a single table. The following extensions record reusable consequences supported by the problem statements.

Corollary 1.6.1. *Two Treatments Give $F = t^2$*
For a one-way normal layout with two treatment groups, the ordinary one-way F statistic equals the square of the pooled two-sample t statistic from Theorem **??**.

Proof. Let the group sizes be n and m, with means $\overline{X}$ and $\overline{Y}$. The between-group sum of squares is

$$SS_B = \frac{nm}{n+m}(\overline{X} - \overline{Y})^2,$$

while $SS_W/(n+m-2)$ is the pooled variance S_p^2. Therefore

$$F = \frac{SS_B}{SS_W/(n+m-2)} = \frac{(\overline{X}-\overline{Y})^2}{S_p^2(n^{-1}+m^{-1})} = t^2.$$

This also matches Corollary **??**, since $t_\nu^2 \sim F_{1,\nu}$ under the null. □

Definition 1.6.2. *Balanced One-Way Random Effects Model*
For $I, J \in \mathbb{N}$, observations $Y_{ij} = \mu + A_i + \varepsilon_{ij}$, $1 \le i \le I$, $1 \le j \le J$, have mutually independent group effects A_i with $\mathbb{E}A_i = 0$ and $\mathrm{Var}(A_i) = \sigma_A^2$, and mutually independent measurement errors ε_{ij} independent of $(A_i)_{i=1}^I$ with $\mathbb{E}\varepsilon_{ij} = 0$ and $\mathrm{Var}(\varepsilon_{ij}) = \sigma_\varepsilon^2$. *Structural Commentary*: $\mathrm{Var}(Y_{ij}) = \sigma_A^2 + \sigma_\varepsilon^2$, and $\mathrm{Cov}(Y_{ij}, Y_{ij'}) = \sigma_A^2$ for $j \neq j'$ within the same group.

Proposition 1.6.3. *Random-Effects Expected Mean Squares*
For Definition 1.6.2 with I groups and J observations per group,

$$\mathbb{E}(MS_W) = \sigma_\varepsilon^2, \qquad \mathbb{E}(MS_B) = \sigma_\varepsilon^2 + J\sigma_A^2.$$

Thus the method-of-moments estimates are

$$\widehat{\sigma}_\varepsilon^2 = MS_W, \qquad \widehat{\sigma}_A^2 = \frac{MS_B - MS_W}{J},$$

when the second quantity is nonnegative.

Proof. Conditional on the random effects, the within-group deviations remove A_i, so the within mean square targets measurement error. Group means have variance $\sigma_A^2 + \sigma_\varepsilon^2/J$, and the between mean square multiplies the excess group-mean variation by J. □

Example 1.6.4. *Dye Strength Batch Variation*
One source problem gives six subsamples from each of six dye batches. From the displayed table,

$$\overline{y}_{\cdot\cdot} = 93.61,$$

and the batch means are approximately

$$92.00,\ 90.50,\ 91.25,\ 98.25,\ 91.17,\ 98.50.$$

The sums of squares are

$$SS_B = 415.47, \qquad SS_W = 134.08.$$

Thus

$$MS_B = 83.09, \qquad MS_W = 4.47,$$

and Proposition 1.6.3 gives

$$\widehat{\sigma}_\varepsilon^2 = 4.47, \qquad \widehat{\sigma}_A^2 = \frac{83.09 - 4.47}{6} = 13.10.$$

In this example, batch-to-batch variation is larger than the within-batch measurement-error component on the variance scale.

Remark 1.6.5. *Factorial Economy and Incomplete Designs*
The source's concluding remarks stress why factorial designs are efficient: the same observations can compare levels of more than one factor and can estimate interactions. As the number of factors grows, full factorial layouts become expensive; fractional factorial designs leave selected cells unobserved while preserving estimability of chosen main effects or interactions. In block experiments, blocks may be too small to contain every treatment, leading to incomplete block designs that retain some blocking advantage without requiring every block-treatment combination.

Remark 1.6.6. *Data-Analysis Problems Need Data Files*
Several source problems refer to external data files such as `tablet`, `LHfemale`, `linseed`, and `fruitfly`. Without those files in the supplied screenshots, the notes can record the intended ANOVA, multiple-comparison, or rank-test workflow, but not the numerical conclusions.